

PAPER_475 — UQFF Sub-Term Physics Modules Catalogue (44 Standalone Calculators)

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Abstract

This whitepaper catalogues 44 standalone sub-term physics calculator modules that constitute the atomic building blocks of the UQFF and MUGE frameworks. Each module encapsulates a single physical quantity — from aether vacuum densities and buoyancy coupling constants to solar cycle frequencies and stellar surface temperatures — as a self-contained C++ class. This catalogue provides a reference index for module equations, key parameter values, output ranges, and integration targets within MAIN_1_CoAnQi.cpp and CondensedPhysics.py.

1. Introduction

The 44 sub-term modules were extracted from grok_share_b0a3dc1d.txt (10,420 lines) and represent the granular physics underpinning the UQFF unified field equation:

$$F_{U,Bi,i} = \int [\text{Ug coupling} \cdot \text{vacuum terms} \cdot \text{buoyancy} \cdot \text{aether} \cdot \text{stellar params}] dx$$

Each module is implemented as a C++ class in modules/subterms/ with standard interface: computeX(), updateVariable(), getEquationText(), printVariables().

2. Module Catalogue

Category A: Vacuum Energy Modules

#	Module	Key Equation	Key Value
1	AetherVacuumDensityModuleA	$\rho_A = E_A/c^2$	$7.09 \times 10^{-36} \text{ J/m}^3$
2	UniversalInertiaVacuumModuleA	[energy]	$7.09 \times 10^{-36} \text{ J/m}^3$
3	ScmVacuumDensityModuleA	$\rho_{vac_SCm} = \rho_{UA}/10$	$7.09 \times 10^{-37} \text{ J/m}^3$
4	UaVacuumDensityModuleA	$\rho_{UA} = E_{UA}/c^2$ [mass]	$7.88 \times 10^{-53} \text{ kg/m}^3$
5	BackgroundAetherModuleA	$\Delta \rho_A = (\rho_A/c^2) \partial_{\mu} \varphi$	$\rho_A = 7.09 \times 10^{-36} \text{ J/m}^3$

Category B: Coupling Constants

#	Module	Key Equation	Key Value
6	AetherCouplingModuleA	$\Delta_{\mu\nu} = g_{\mu\nu} + \eta T_s^{\mu\nu}$	$\eta \approx 1/E_s_total \approx 1e-15$

#	Module	Key Equation	Key Value
7	UgCouplingModule	$k_i \text{ weights } U_{g_i}$	$k_1=1.5, k_2=1.2, k_3=1.8, k_4=1.0$
8	BuoyancyCouplingModule	$\bar{U}_{bi} = -\beta_i U_{gi} \Omega_{g...}$	$\beta_i = 0.6$ uniform
9	InertiaCouplingModule	$\bar{U}_e = M r^2 \Omega^2/c^2$	Sun: $\approx 7.8 \times 10^{-10}$
10	UgIndexModule	$U_{g_{\{i,n\}}} = G M/r^2 Q_{i...}$	4×26 array

Category C: Solar/Stellar Parameters

#	Module	Key Equation	Key Value
11	SolarWindBuoyancyModule	$\epsilon_{sw} = 0.001 \times (1 + A \sin(\omega t))$	$\epsilon_{sw} = 0.001$ baseline
12	SolarWindModulationModule	$v(t) = v_0(1+A \sin(\omega t))$	$v_0=400$ km/s, $A=0.2$
13	SolarWindVelocityModule	$v_{sw} \in [400,800]$ km/s	$v_{fast} = 750$ km/s
14	SolarCycleFrequencyModule	$f_{cyc} = 1/(11 \text{ yr})$	2.88×10^{-9} Hz
15	HeliosphereThicknessModule	$r_{HP} - r_{TS}$	~ 30 AU ($\sim 4.5 \times 10^{12}$ m)
16	StellarMassModule	$M_s(t) = M_0(1-\gamma_{ML} t)$	$\gamma_{ML} \approx 1.4 \times 10^{-14} \text{ s}^{-1}$
17	StellarRotationModule	$\omega_s = 2\pi/P_{rot}$	Sun: 2.87×10^{-6} rad/s
18	SurfaceMagneticFieldModule	$B_{pole} = M_{mag}/4\pi r^3$	Magnetar: 4.4×10^{13} T
19	SurfaceTemperatureModule	$T_{eff} = (L/4\pi\sigma r^2)^{0.25}$	Sun: 5778 K
20	MagneticMomentModule	$\mu = I A_{vort}$	Solar: $\sim 1 \times 10^{21} \text{ A}\cdot\text{m}^2$

Category D: Galactic/Astrophysical Parameters

#	Module	Key Equation	Key Value
21	GalacticDistanceModule	$L_g = \text{virial radius}$	MW: 2.55×10^{20} m
22	GalacticSpinModule	$\Omega_g = 2\pi/T_{gal}$	MW: 7.3×10^{-16} rad/s
23	GalacticBlackHoleModule	$M_{BH} \propto \sigma^4 (M-\sigma)$	Sag A*: 8×10^{36} kg
24	FeedbackFactorModule	$f_{env} = f_{AGN} + f_{SN} + f_{SF}$	$f_{AGN}=0.1, f_{SN}=0.05$
25	MagneticStringModule	$T_s = (\mu_0 I^2/4\pi) \ln(L/a)$	$\sim 10^{28}$ N (cosmic string)
26	Ug3DiskVectorModule	$U_{g3_disk} = G M_{disk}/r^2(h/r)$	MW: $h/r \approx 0.07$
27	Ug1DefectModule	$U_{g1_corr} = U_{g1}(1-\delta_{def})$	$\delta_{def} \approx 0.05-0.15$

Category E: Quantum & Field Calculators

#	Module	Key Equation	Key Value
28	DPMModule	26-sphere: $(x-h)^2 + \dots = r^2$	SCm $E = 10^{42}$ J
29	UnifiedFieldModule	$F_U = \sum k_i U_{g_i} + F_{bi...}$	Orchestrator
30	StressEnergyTensorModule	$T_{\mu\nu} = (\rho+p)u_\mu u_\nu + pg_{\mu\nu}$	Trace = $-\rho+3p$
31	QuasiLongitudinalModule	$\bar{E}_{QL} = \epsilon_0 E^2/2$	$\sim 1e-12 \text{ J/m}^3$
32	OuterFieldBubbleModule	$r_0 \exp(H t)$	At 10 Gyr: $r = 1.28 r_0$
33	ReciprocationDecayModule	$r_0 \exp(-t/\tau_{rec})$	$\tau_{rec} \sim 1$ Gyr
34	ScmPenetrationModule	$\delta_{SCm} = \lambda(\rho/\rho_{SCm})^{0.5}$	London-analog depth
35	ScmReactivityDecayModule	$d[SCm]/dt = -k_r [SCm]$	[SSq]=0.57, $k_r \sim 1e-18 \text{ s}^{-1}$
36	ScmVelocityModule	$v_{SCm} = c/n_{SCm}$	$n_{SCm} \approx 1.0000001$
37	PiConstantModule	$\pi = 4 \sum (-1)^k / (2k+1)$	Leibniz series
38	CorePenetrationModule	$\delta = (\rho_{core}/\rho_{avg})^n r_{core}$	NS core: $\delta \rightarrow 0$
39	NegativeTimeModule	$g(t<0) = g_0 \cos(\omega t)$	$f_{TRZ} = 0.1$

#	Module	Key Equation	Key Value
40	TimeReversalZoneModule	$r_{TRZ} = r_{TRZ}/r$	Canonical: 0.1
41	HeavisideFractionModule	$f(r)$ (f<0), 1 (f≥0)	Threshold: 0
42	StepFunctionModule	$\theta(x)$: boxcar windows	Regime switching

Category F: Index / Utility Modules

#	Module	Key Equation	Key Value
43	GalacticBlackHoleModule	$\mu_{BH} = G M_{BH}/r^2$	At $r=5.5 \times 10^{10}$ m (Sag A*)
44	InertiaCouplingModule	$\gamma_d = 1 + I_c$	Ug3 relativistic boost

3. Integration Targets

3.1 MAIN_1_CoAnQi.cpp

Sub-term modules are candidates for batch extraction as PhysicsTerm subclasses:

```
// Example: SCmVacuumDensityModule → SCmVacuumDensityTerm
class SCmVacuumDensityTerm : public PhysicsTerm {
public:
    double compute(const SystemParams& params) override {
        return 7.09e-37; // ρ_vac_SCm [J/m³]
    }
    std::string getName() const override { return "SCm Vacuum Density"; }
};
```

Target integration point: **Batch 24** in SOURCE1-116 registry near line 6688+.

3.2 CondensedPhysics.py

Each sub-term module maps to a Calculator class method:

```
class SubTermCalculator:
    def compute_aether_vacuum_density(self) -> dict:
        # Returns ρ_vac_A = 7.09e-36 J/m³ with equation text
```

Target: new UQFFSubTermCalculator in CondensedPhysics.py Part 5 (after line 60000).

3.3 index.js

Module constants exportable as:

```
const SUB_TERMS = {
    rho_vac_UA: 7.09e-36, // J/m³
    rho_vac_SCm: 7.09e-37, // J/m³
    beta_i: 0.6, // buoyancy coupling
    kappa: 0.0005/86400, // per-second DPM rate
    SSq: 0.57, // SCm calibration
};
```

4. Calibration Constants Summary

Constant	Value	Source Module
κ	0.0005/day	DPMModule
[SSq]	0.57	ScmReactivityDecayModule
β_i	0.6 (all i)	BuoyancyCouplingModule
ε_{sw}	0.001	SolarWindBuoyancyModule
η	$\sim 10^{-15}$	AetherCouplingModule
H_SCm	0.99	ScmVacuumDensityModule
k_1, k_2, k_3, k_4	1.5/1.2/1.8/1.0	UgCouplingModule
f_TRZ	0.1	TimeReversalZoneModule

5. Scientific Context

The 44 sub-term modules collectively represent a field-theoretic decomposition of gravity that accounts for:

- **Vacuum energy hierarchy:** $\rho_{UA} : \rho_{SCm} : \rho_{cosm} = 10 : 1 : 0.001$
- **Coupling hierarchy:** k_3 (string/rotation) $> k_1$ (dipole) $> k_2$ (charge) $> k_4$ (vacuum)
- **Time evolution:** DPM birth model $\rightarrow$ inflation $\rightarrow$ SCm decay $\rightarrow$ solar wind modulation $\rightarrow$ present
- **Multi-scale validity:** Sub-parsec (heliosphere) to Gpc (galaxy clusters)

6. Conclusion

This catalogue documents 44 atomic physics calculator modules that implement the sub-term physics of the UQFF/MUGE framework. The modules span vacuum energy, coupling constants, stellar parameters, galactic scales, and quantum field calculators. They constitute the modules/subterms/ library and are ready for integration as PhysicsTerm subclasses in MAIN_1_CoAnQi.cpp or as Python calculator methods in CondensedPhysics.py.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see uqff_lagrangian_derivation.py).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_{NS})(\partial^\mu \phi_{NS}) - V(\phi_{NS}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2) \phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.106 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 101, \quad n_{\text{channel}} = 8/26$$

Since $p_{\text{DVP}} = 101$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10⁴ yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.106	✓ Threshold-consistent
DVP prime BSH layers	$p_k \in \{2,3,\dots,113\}$ 26 harmonic terms	$p_{\text{DVP}} = 101$ $j = 1\dots 26,$ $\cos(2\pi j/26)$	✓ Resonant ✓ Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_{T} (QED synchrotron)	UQFF U_{m} scattering kernel: $\sigma_{\text{T}} = 6.6524 \times 10^{-29} \text{ m}^2$	$\sigma_{\text{T}} = 6.6524 \times 10^{-29} \text{ m}^2$ (PDG QED exact)	PDG 2024	100% (exact QED input)
Astrophysical system luminosity X-ray / Radio	UQFF MUGE $g_{\text{total}} \rightarrow L_{\text{X}}$ via Stefan-Boltzmann + buoyancy flux: $L_{\text{X}} \approx g_{\text{total}} \times M_{\text{env}}$	$L_{\text{X}} \geq 10^{37} \text{ erg/s}$	Chandra CXC	✓ Consistent order of magnitude
GR Schwarzschild limit	UQFF g_{total} must satisfy $g \leq c^2/(2r_{\text{s}})$ at event horizon	$r_{\text{s}} = 2GM/c^2$ (GR exact)	PDG 2024 / GR	✓ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$	Observed X-ray variability τ_{obs} (instrument monitoring)	Chandra CXC	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Astrophysical system through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

Source: grok_share_b0a3dc1d.txt L1502-9356 (44 class definitions)

Header index: modules/subterms/sub_terms_index.h

Tags: sub-terms, catalogue, vacuum-density, coupling-constants, solar-parameters, UQFF, MUGE, integration